20261030 指标修复说明

datadate	rdq	jdate	date	fyearq	fqtr
财务期间截止日（Compustat）	季度盈利公告日（Compustat）	形成月末（代码生成）	收益月末（代码生成）	公司财年（Compustat）	财年内季度（Compustat）

一、SUE

形成月与公告日

修复前 41.4256%（938,820 / 2,266,282） **修复后** 0%（0 / 2,473,877）
代码 `chars_ciz/sue.py` **参考** HXZ（公告日匹配）；JKP（公司匹配）

原代码

```
SELECT a.gvkey, a.permno, a.datadate, b.date, a.sue
FROM ccm2 a
LEFT JOIN crsp_msf b
    ON a.datadate <= b.date
    AND a.plus12m >= b.date
ORDER BY a.permno, b.date, a.datadate DESC
```

修复代码

```
SELECT a.gvkey, b.permno, a.datadate, b.date,
↪  a.sue,
       a.fyearq, a.fqtr, a.rdq
FROM (SELECT * FROM crsp_msf
      WHERE date >= DATE '1972-01-01') b
LEFT JOIN ccm2 a
    ON a.gvkey = b._current_gvkey
    AND a.rdq < b.date
    AND a.plus6m >= b.date
ORDER BY b.permno, b.date, a.datadate DESC, a.rdq
↪  DESC, a.fyearq DESC, a.fqtr DESC
```

HXZ A.1.1，p.2072（PDF 54）：SUE 同比、历史期数与公告日

A.1.1 Sue1, Sue6, and Sue12, standardized unexpected earnings. Per Foster, Olsen, and Shevlin (1984), Sue denotes Standardized Unexpected Earnings, and is calculated as the change in split-adjusted quarterly earnings per share (Compustat quarterly item EPSPXQ divided by item AJEXQ) from its value four quarters ago divided by the standard deviation of this change in quarterly earnings over the prior eight quarters (six quarters minimum). At the beginning of each month t, we split all NYSE, Amex, and NASDAQ stocks into deciles based on their most recent past Sue. Before 1972, we use the most recent Sue computed with quarterly earnings from fiscal quarters ending at least four months prior to the portfolio formation. Starting from 1972, we use Sue computed with quarterly earnings from the most recent quarterly earnings announcement dates (Compustat quarterly item RDQ). For a firm to enter our portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Sue to be within six months prior to the portfolio formation. We do so to exclude stale information on earnings. To avoid potentially erroneous records, we also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly portfolio returns are calculated, separately, for the current month t

1972 年起按 `rdq` 匹配，此前用四个月滞后；同比相隔 12 个月，标准差取过去八期中的至少六项。

$$\mathrm{SUE}_q = \frac{\Delta EPS_q}{\mathrm{sd}(\Delta EPS_{q-1}, \ldots, \Delta EPS_{q-8})}, \qquad \Delta EPS_q = EPS_q - EPS_{q-4}, \quad EPS_q = \frac{\texttt{epspxq}_q}{\texttt{ajexq}_q}.$$

			Before			After			
date	jdate	permno	fyearq / fqtr	datadate	sue	fyearq / fqtr	datadate	rdq	sue
2023-03-31	2023-02-28	10026	2023 / 1	2022-12-31	-0.241219	2023 / 1	2022-12-31	2023-01-30	-0.241219
2023-04-30	2023-03-31	10026	2023 / 2	2023-03-31	0.214297	2023 / 1	2022-12-31	2023-01-30	-0.241219
2023-05-31	2023-04-30	10026	2023 / 2	2023-03-31	0.214297	2023 / 1	2022-12-31	2023-01-30	-0.241219
2023-06-30	2023-05-31	10026	2023 / 2	2023-03-31	0.214297	2023 / 2	2023-03-31	2023-05-01	0.214297
2023-03-31	2023-02-28	10200	2022 / 4	2022-12-31	3.910065	2022 / 4	2022-12-31	2023-02-22	3.910065
2023-04-30	2023-03-31	10200	2023 / 1	2023-03-31	-6.098537	2022 / 4	2022-12-31	2023-02-22	3.910065
2023-05-31	2023-04-30	10200	2023 / 1	2023-03-31	-6.098537	2022 / 4	2022-12-31	2023-02-22	3.910065
2023-06-30	2023-05-31	10200	2023 / 1	2023-03-31	-6.098537	2023 / 1	2023-03-31	2023-05-02	-6.098537

年季合并：gvkey 不同或缺失

修复前（SUE 漏接） 0.1243%（2,402 / 1,932,512） **修复后** 0%（0 / 1,932,512）
代码 `chars_ciz/impute_rank_output.py` **参考** JKP（先核对当月公司，再合并年季指标）

原代码

```
df = df_a_sel.join(
    df_q_sel, on=['gvkey', 'permno', 'jdate'],
    how='left'
)
```

修复代码

```
df = df_a_sel.join(
    df_q_sel, on=['permno', 'jdate'],
    how='left', validate='1:1'
)
```

						sue	
permno	jdate	date	年度 gvkey	季度 gvkey	季度源值	Before	After
55335	1984-07-31	1984-08-31	005207	001066	0.328600	NULL	NULL
16851	2018-01-31	2018-02-28	NULL	004060	-0.165614	NULL	-0.165614

二、RE

历史月份错配

`statpers`：预测快照日；`fpedats`：预测目标财年截止日；`meanest`：该财年 EPS 预测均值。

修复前 84.5309%（2,228,395 / 2,636,190） **修复后** 0%（0 / 1,306,959）
代码 `chars_ciz/myre.py` **参考** HXZ（逐月预测变化及六个月汇总）

原代码

```
.unique(...)
.shift(1).over('permno_fpedats')
```

修复代码

```
ON a.ticker=b.ticker AND a.permno=b.permno AND a.fpedats=b.fpedats
AND b.month_id=a.month_id-1

ON a.ticker=b.ticker AND a.permno=b.permno
AND b.month_id=a.month_id-{k}
```

$$\mathrm{RE}_t = \sum_{k=0}^{5} \mathrm{revision}_{t-k}, \qquad \mathrm{revision}_t = \frac{\mathtt{meanest}_t - \mathtt{meanest}_{t-1}}{\mathtt{prc_adj}_t}, \qquad \mathtt{prc_adj}_t = \frac{\mathtt{prc}_{t-1}}{\mathtt{cfacpr}_{t-1}}.$$

Panel A：Before

permno	statpers	lag_date_1	lag_date_2	lag_date_3	lag_date_4	lag_date_5	lag_date_6	re
14392	2014-10-16	2014-11-20	2014-12-18	2014-09-18	NULL	NULL	NULL	-0.000857
14423	2020-07-16	2020-09-17	2019-04-18	2020-02-20	NULL	NULL	NULL	-0.007797
14449	2016-07-14	2016-02-18	2015-04-16	2016-11-17	NULL	NULL	NULL	0.106589

Panel B：After

permno	statpers	lag_date_1	lag_date_2	lag_date_3	lag_date_4	lag_date_5	lag_date_6	re
14392	2014-10-16	2014-10-16	2014-09-18	2014-08-14	2014-07-17	2014-06-19	2014-05-15	-0.000671
14423	2020-07-16	2020-07-16	2020-06-18	2020-05-14	2020-04-16	2020-03-19	NULL	-0.007377
14449	2016-07-14	2016-07-14	2016-06-16	2016-05-19	2016-04-14	2016-03-17	NULL	-0.004702

			re	
date	jdate	permno	Before	After
2014-11-30	2014-10-31	14392	-0.000857	-0.000671
2020-08-31	2020-07-31	14423	-0.007797	-0.007377
2016-08-31	2016-07-31	14449	0.106589	-0.004702

换财年后保留此前 FY1 修正

修复前 25.8587%（289,322 / 1,118,858） **修复后** 0%（0 / 1,118,858）
代码 `chars_ciz/myre.py` **参考** HXZ（按月定义推导）

原代码

```
ON a.ticker=b.ticker AND a.permno=b.permno AND
↪  a.fpedats=b.fpedats
AND b.month_id=a.month_id-{k}
```

修复代码

```
ON a.ticker=b.ticker AND a.permno=b.permno
AND b.month_id=a.month_id-{k}
```

FY1 由 FY2022 切换到 FY2023；2022-10 只有 FPI=2 记录。

预测数据及月度计算结果（价格来源调整前）

permno	month	statpers	fpi	fpedats	meanest	meanest_last_month	prc_adj	monthly_revision
10026	2022-06	2022-06-16	1	2022-09-30	2.73	2.74	128.22	-0.000077991
10026	2022-07	2022-07-14	1	2022-09-30	2.74	2.73	139.66	0.000071602
10026	2022-08	2022-08-18	1	2022-09-30	2.59	2.74	135.51	-0.001106929
10026	2022-09	2022-09-15	1	2022-09-30	2.59	2.59	149.03	0.000000000
10026	2022-10	2022-10-20	2	2023-09-30	4.87	–	–	–
10026	2022-11	2022-11-17	1	2023-09-30	4.81	NULL	147.61	NULL
10026	2022-12	2022-12-15	1	2023-09-30	4.83	4.81	164.02	0.000121936

Panel A：Before

permno	statpers	fpedats	revision_l0	revision_l1	revision_l2	revision_l3	revision_l4	revision_l5	re
10026	2022-11-17	2023-09-30	NULL	NULL	NULL	NULL	NULL	NULL	NULL
10026	2022-12-15	2023-09-30	0.000121936	NULL	NULL	NULL	NULL	NULL	NULL

Panel B：After

permno	statpers	fpedats	revision_l0	revision_l1	revision_l2	revision_l3	revision_l4	revision_l5	re
10026	2022-11-17	2023-09-30	NULL	NULL	0.000000000	-0.001106929	0.000071602	-0.000077991	-0.001113
10026	2022-12-15	2023-09-30	0.000121936	NULL	NULL	0.000000000	-0.001106929	0.000071602	-0.000913

			re	
date	jdate	permno	Before	After
2022-12-31	2022-11-30	10026	NULL	-0.001113
2023-01-31	2022-12-31	10026	NULL	-0.000913

FY1 与 FY2 的预测期限

`fpi=1` 是最近尚未公布全年业绩的财年（FY1）；`fpi=2` 是其后一年（FY2）。

FY2 对应更远一期盈利，与 FY1 混用会改变信号期限；10 月 FY2 与 11 月 FY1 虽同指 FY2023，也不代表预测期限相同。

HXZ（2020）附录 A.1.3，第 2073 页（两处原文短片段）

for firm i's current fiscal year earnings (fiscal period indicator = 1),

require a minimum of four monthly forecast changes when constructing Re.

修复前 50.5170%（1,753,844 / 3,471,788） **修复后** 0%（0 / 1,717,944）
代码 `chars_ciz/myre.py` **参考** HXZ（FY1 预测）

原代码

```
pl.col('medest').is_not_null() &
↪  pl.col('fpedats').is_not_null()
```

修复代码

```
(pl.col('fpi') == '1') &
```

$$\texttt{monthly_revision} = \frac{\texttt{meanest} - \texttt{meanest_last_month}}{\texttt{prc_adj}}$$

					Before		After	
permno	statpers	fpi	meanest	prc_adj	meanest_ last_month	monthly_ revision	meanest_ last_month	monthly_ revision
10026	2022-06-16	2	4.78	128.22	4.73	0.000390	–	–
10026	2022-07-14	2	4.88	139.66	4.78	0.000716	–	–
10026	2022-08-18	2	4.85	135.51	4.88	-0.000221	–	–
10026	2022-09-15	2	4.85	149.03	4.85	0.000000	–	–
10026	2022-10-20	2	4.87	129.47	4.85	0.000154	–	–
10026	2022-11-17	1	4.81	147.61	4.87	-0.000406	NULL	NULL
10026	2022-12-15	1	4.83	164.02	4.81	0.000122	4.81	0.000122

After 中，–：FPI=2，已筛除；NULL：记录保留，但没有上月 FY1 预测。

汇总月份、有效项数与求和

输出变化 96.4680%（1,449,467 / 1,502,537） **公式检查** 0%（0 / 1,306,959）
代码 `chars_ciz/myre.py` **参考** HXZ（附录 A.1.3，公式 A2）

调整前代码

```
range(1, 7)
len() >= 3
list.mean()
```

修复代码

```
range(0, 6)
len() >= 4
list.sum()
```

预测数据及月度计算结果（价格来源调整前）

permno	statpers	fpedats	meanest	previous_statpers	prc_adj	monthly_revision
10026	2022-09-15	2022-09-30	2.59	2022-08-18	149.03	0.000000000
10026	2022-11-17	2023-09-30	4.81	NULL	147.61	NULL
10026	2022-12-15	2023-09-30	4.83	2022-11-17	164.02	0.000121936
10026	2023-01-19	2023-09-30	4.80	2022-12-15	149.71	-0.000200387
10026	2023-02-16	2023-09-30	4.11	2023-01-19	143.30	-0.004815073
10026	2023-03-16	2023-09-30	4.11	2023-02-16	141.21	0.000000000
10026	2023-04-20	2023-09-30	4.06	2023-03-16	148.22	-0.000337336
10026	2023-05-18	2023-09-30	3.91	2023-04-20	153.20	-0.000979112

Panel A：Before

permno	statpers	fpedats	revision_l0	revision_l1	revision_l2	revision_l3	revision_l4	revision_l5	revision_l6	re
10026	2023-03-16	2023-09-30	–	-0.004815073	-0.000200387	0.000121936	NULL	NULL	0.000000000	-0.001223

Panel B：After

permno	statpers	fpedats	revision_l0	revision_l1	revision_l2	revision_l3	revision_l4	revision_l5	revision_l6	re
10026	2023-03-16	2023-09-30	0.000000000	-0.004815073	-0.000200387	0.000121936	NULL	NULL	–	-0.004894

			re	
date	jdate	permno	Before	After
2023-02-28	2023-01-31	10026	-0.000228	-0.001185
2023-03-31	2023-02-28	10026	-0.000296	-0.004894
2023-04-30	2023-03-31	10026	-0.001223	-0.004894
2023-05-31	2023-04-30	10026	-0.001223	-0.005231
2023-06-30	2023-05-31	10026	-0.001046	-0.006210

HXZ A.1.3 / A2：取当月和前五个月，至少四项，按公式求和。上表单独比较汇总规则，保留调整前的 CRSP 价格；价格来源的后续修正见下表。– 表示窗口外。

EPS、价格来源与拆股调整

修复前 调整后 IBES EPS、CRSP 调整价 **修复后** IBES 原始 EPS、上一月原始 PRICE
代码 `chars_ciz/myre.py`；`download_data.py` **参考** HXZ A.1.3 / A2；WRDS 拆股说明 Method 3

调整体系不同会把拆股误算为预测变化。令 f, f_0 为当月、上月原始预测，p_0 为上月原始价格；c, c_0 为预测日股数因子，c_{p0} 为价格日股数因子：

$$\text{monthly_revision} = \frac{f - f_0 c/c_0}{p_0 c/c_{p0}}.$$

JJSF（10026）：分母及月度变化

statpers	meanest change	prc_adj (Before)	PRICE (After)	prdays (After)	monthly_revision (Before / After)
2022-12-15	4.83 − 4.81	164.02	156.08	2022-11-16	0.000121936 / 0.000128139
2023-01-19	4.80 − 4.83	149.71	156.04	2022-12-14	−0.000200387 / −0.000192258
2023-02-16	4.11 − 4.80	143.30	148.41	2023-01-18	−0.004815073 / −0.004649282

PRICE 为原始股价，prdays 为报价日；本例无拆股。

`chars_raw_no_impute.parquet`：价格来源调整前后

date	jdate	permno	re (Before)	re (After)
2023-02-28	2023-01-31	10026	-0.001185	-0.001312
2023-03-31	2023-02-28	10026	-0.004894	-0.004855
2023-04-30	2023-03-31	10026	-0.004894	-0.004855

HXZ A.1.3 / A2，p.2073（PDF 55）：FY1、原始 EPS / PRICE、拆股及至少四项

A.1.3 Re1, Re6, and Re12, revisions in analyst earnings forecasts. Following Chan, Jegadeesh, and Lakonishok (1996), we measure earnings surprise as the revisions in analysts' forecasts of earnings obtained from the Institutional Brokers' Estimate System (IBES). Because analysts' forecasts are not necessarily revised each month, we construct a 6-month moving average of past changes in analysts' forecasts:

$$\text{Re}_{it} = \sum_{\tau=1}^{6} \frac{f_{it-\tau} - f_{it-\tau-1}}{p_{it-\tau-1}}, \tag{A2}$$

in which $f_{it-\tau}$ is the consensus mean forecast (unadjusted IBES file, item MEANEST) issued in month $t-\tau$ for firm i's current fiscal year earnings (fiscal period indicator = 1), and $p_{it-\tau-1}$ is the prior month's share price (unadjusted file, item PRICE). We require both earnings forecasts and share prices to be in U.S. dollars (currency code = USD). We also adjust for any stock splits and require a minimum of four monthly forecast changes when constructing Re. At the beginning

WRDS（2006）Method 3，PDF 6：以 CRSP 日度股数因子统一每股基础

Method 3

Perhaps, relatively the most reliable and accurate way of joining Unadjusted Detail History and Actuals is to use CRSP cumulative adjustment split factor extracted from the CRSP Daily files as it contains more precise information regarding the true split date of a stock compared with IBES split factor . This method involves the following steps:

三、ROA

按盈利公告日匹配季度值

修复前 0.9096%（24,183 / 2,658,558） **修复后** 0%（0 / 2,494,209）
代码 `chars_ciz/accounting.py` **参考** HXZ A.4.3

上框统计非空季度 ROA 距季末超过六个月的比例。

原代码

```
pl.col('datadate').dt.offset_by('4mo').dt.month_end().alias('jdate')
```

修复代码（节选）

```
roa_events = roa_reports.filter(pl.col('rdq') > pl.col('datadate'))
# ... 同公司匹配上季资产，得到 atq_l1 ...
roa_events = roa_events.with_columns(
    (pl.col('ibq') / pl.col('atq_l1').replace(0, None)).alias('roa'))
roa_public = (pl.concat([
    roa_events.with_columns(pl.col('datadate').dt.offset_by(f'{k}mo').dt.month_end().alias('jdate'))
    for k in range(1, 7)])
    .filter(pl.col('rdq') < pl.col('jdate'))
    .sort(['gvkey', 'jdate', 'rdq', 'datadate'], descending=[False, False, True, True])
    .unique(subset=['gvkey', 'jdate'], keep='first', maintain_order=True))
```

$$\mathrm{ROA}_q = \frac{\texttt{ibq}_q}{\texttt{atq}_{q-1}}.$$

HXZ A.4.3，p.2093（PDF 75）：ROA 分母为上季度资产，按公告日使用

A.4.3 Roa1, Roa6, and Roa12, return on assets. Return on assets, Roa, is income before extraordinary items (Compustat quarterly item IBQ) divided by 1-quarter-lagged total assets (item ATQ). At the beginning of each month t, we sort stocks into deciles on Roa computed with quarterly earnings from the most recent earnings announcement dates (item RDQ). For a firm to enter the portfolio formation, we require the end of the fiscal quarter that corresponds to its most recent Roa to be within six months prior to the portfolio formation. We also require the earnings announcement date to be after the corresponding fiscal quarter end. Monthly decile returns are calculated for

Panel A：Before，2023-01-31 形成月

permno	datadate	ibq	atq_l1	roa
10026	2022-09-30	17.310	1291.242	0.013406

Panel B：After，同一形成月

permno	datadate	ibq	atq_l1	roa
10026	2022-12-31	6.633	1216.966	0.005450

			Before		After	
date	jdate	permno	roa (Quarterly)	roa (Final)	roa (Quarterly)	roa (Final)
2023-02-28	2023-01-31	10026	0.013406	0.013406	0.005450	0.005450
2023-03-31	2023-02-28	10026	0.013406	0.013406	0.005450	0.005450
2023-04-30	2023-03-31	10026	0.013406	0.013406	0.005450	0.005450
2023-05-31	2023-04-30	10026	0.005450	0.005450	0.005450	0.005450
2023-06-30	2023-05-31	10026	0.005450	0.005450	0.005597	0.005597

HXZ A.4.3：使用最近公告日（RDQ）的盈利；季末距形成月不超过六个月，公告日晚于季末。

分母来自同公司、对应期间

代码	`chars_ciz/accounting.py`
参考	HXZ（ROA 公式）；GHZ（公司首期缺历史时置空）；JKP（公司内、按月匹配）

原代码

```
pl.col('atq').shift(1).over('permno')
```

修复代码

```
calendar_lag(frame, asset, 1, months, history=raw)
```

$$\mathrm{ROA}_q = \frac{\texttt{ibq}_q}{\texttt{atq}_{q-1}}, \qquad \mathrm{ROA}_y = \frac{\texttt{ib}_y}{\texttt{at}_{y-1}}.$$

同一 gvkey 的上季/上年资产；无记录、资产为零或同期数值冲突时为 NULL。

MULN：季度 ROA 计算对照

				atq_l1		roa	
permno	gvkey	datadate	ibq	Before	After	Before	After
12350	040116	2021-12-31	-156.062	30.773	NULL	-5.071394	NULL
12350	040116	2022-03-31	-324.608	45.413	45.413	-7.147909	-7.147909

新公司首季缺本公司上季资产，ROA 为 NULL；后续恢复计算。

年度 ROA：缺失年度不得回退到更早年度

				at_{y-1}		roa	
permno	gvkey	datadate	ib	Before	After	Before	After
10066	014112	1999-06-30	-2.413	7.253	8.892	-0.332690	-0.271368

修复前按行 `shift(1)`，跳过 FY1990–FY1998 后使用 FY1989 资产；修复后取同一 gvkey 的 FY1998 资产。

四、AGR / CHTX / NI / CHCSHO

财务滞后统一用 `calendar_lag`：按同一股票、gvkey 和目标年月连接。

```
calendar_lag(data_rawq, field, k, months=3)
calendar_lag(data_rawa, field, k, months=12)
```

匹配去年同季

指标	修复前	修复后
AGR	3.0731%（76,272 / 2,481,899）	0%（0 / 2,265,450）
CHCSHO	2.9703%（72,861 / 2,452,944）	0%（0 / 2,260,830）
CHTX	3.0309%（74,691 / 2,464,334）	0%（0 / 2,250,289）
NI	14.2624%（398,141 / 2,791,546）	0%（0 / 2,260,803）
代码	`chars_ciz/accounting.py` **参考**	HXZ、GHZ（指标定义）；JKP（同期匹配）

原代码

```
.shift(4).over('permno')
```

修复代码

```
b.period_id=a.period_id-12 AND b.rdq<a.jdate
```

$$\text{AGR} = \frac{\texttt{atq}_q - \texttt{atq}_{q-4}}{\texttt{atq}_{q-4}}, \qquad \text{CHTX} = \frac{\texttt{txtq}_q - \texttt{txtq}_{q-4}}{\texttt{atq}_{q-4}}, \qquad \text{CHCSHO} = \frac{\texttt{cshoq}_q}{\texttt{cshoq}_{q-4}} - 1.$$

$$\text{NI} = \ln(\texttt{cshoq}_q \times \texttt{ajexq}_q) - \ln(\texttt{cshoq}_{q-4} \times \texttt{ajexq}_{q-4}).$$

2013 年 2—4 月：财务数据与指标计算

Panel A：Before

permno	Compustat	datadate	fyearq / fqtr	rdq	atq	txtq	cshoq	ajexq
10028	Current	2012-09-30	2012 / 3	2012-11-13	24.489	0.028	12.176	1.0
10028	Lag	2010-12-31	2010 / 4	2011-04-15	30.260	0.020	9.986	1.0

Panel B：After

permno	Compustat	datadate	fyearq / fqtr	rdq	atq	txtq	cshoq	ajexq
10028	Current	2012-09-30	2012 / 3	2012-11-13	24.489	0.028	12.176	1.0
10028	Lag	2011-09-30	2011 / 3	2011-11-14	42.495	0.000	11.220	1.0

指标	修复前	修复后
AGR	$(24.489 - 30.260)/30.260 = -0.190714$	$(24.489 - 42.495)/42.495 = -0.423720$
CHCSHO	$12.176/9.986 - 1 = 0.219307$	$12.176/11.220 - 1 = 0.085205$
CHTX	$(0.028 - 0.020)/30.260 = 0.000264$	$(0.028 - 0)/42.495 = 0.000659$
NI	$\ln(12.176/9.986) = 0.198283$	$\ln(12.176/11.220) = 0.081769$

				agr		chcsho		chtx		ni	
date	jdate	datadate	permno	Before	After	Before	After	Before	After	Before	After
2013-01-31	2012-12-31	2011-09-30	10028	0.508198	0.508198	0.137470	0.137470	0.000000	0.000000	0.128806	0.128806
2013-02-28	2013-01-31	2012-09-30	10028	-0.190714	-0.423720	0.219307	0.085205	0.000264	0.000659	0.198283	0.081769
2013-03-31	2013-02-28	2012-09-30	10028	-0.190714	-0.423720	0.219307	0.085205	0.000264	0.000659	0.198283	0.081769
2013-04-30	2013-03-31	2012-09-30	10028	-0.190714	-0.423720	0.219307	0.085205	0.000264	0.000659	0.198283	0.081769
2013-05-31	2013-04-30	2012-12-31	10028	-0.160667	-0.055447	0.219307	0.000987	0.001736	0.000506	0.198283	0.000986

年度：缺失年度不得回退

permno 10066（gvkey 014112）：FY1999 年度指标

Panel A：Before					
项目	datadate	at	txt	csho	ajex
当期	1999-06-30	9.435	0.007	25.797	0.01428571
滞后	1989-06-30	7.253	0.339	59.095	0.01428571
Panel B：After					
项目	datadate	at	txt	csho	ajex
当期	1999-06-30	9.435	0.007	25.797	0.01428571
滞后	1998-06-30	8.892	0.003	18.344	0.01428571

指标	Before	After
AGR	$(9.435-7.253)/7.253=0.300841$	$(9.435-8.892)/8.892=0.061066$
CHTX	$(0.007-0.339)/7.253=-0.045774$	$(0.007-0.003)/8.892=0.000450$
CHCSHO	$25.797/59.095-1=-0.563466$	$25.797/18.344-1=0.406291$
NI	$\ln(25.797/59.095)=-0.828888$	$\ln(25.797/18.344)=0.340956$

修复前按行 `shift(1)`，跳过 FY1990–FY1998 后使用 FY1989；修复后按同一 gvkey 的 `period_id`−12 取 FY1998。

NI：缺少历史股数不得用零替代

季度修复前 11.6199%（324,375 / 2,791,546）　　**季度修复后** 0%（0 / 2,260,803）
年度变化 8.2413%（307,163 / 3,727,137）；空值变化或数值差大于 10^{-10}。
代码 `chars_ciz/accounting.py`　**参考** HXZ（Nsi 年度公式，扩展至季度）；JKP（同期匹配）

原代码

```
.log().fill_nan(0).fill_null(0)
```

修复代码

```
calendar_lag(data_rawa, c, 1, 12,
             history=annual_ni_history)
pl.when((pl.col('csho') * pl.col('ajex') > 0) &
        (pl.col('_ni_csho_l1') * pl.col('_ni_ajex_l1') > 0)).then(...)
pl.when((pl.col('cshoq') * pl.col('ajexq') > 0) &
        (pl.col('cshoq_l4') * pl.col('ajexq_l4') > 0)).then(...)
```

HXZ A.3.10，p.2088（PDF 70）：NI 为调整后股数比值的对数

A.3.10 Nsi, net stock issues. At the end of June of year t, we measure net stock issues, Nsi, as the natural log of the ratio of the split-adjusted shares outstanding at the fiscal year ending in calendar year $t-1$ to the split-adjusted shares outstanding at the fiscal year ending in $t-2$. The split-adjusted shares outstanding is shares outstanding (Compustat annual item CSHO) times the adjustment factor (item AJEX). At the end of June of each year t, we sort stocks with negative Nsi

$$\text{NI}=\ln(\text{csho}_t\times\text{ajex}_t)-\ln(\text{csho}_{t-1}\times\text{ajex}_{t-1}).$$

年度使用上年数据；季度使用去年同季数据。

年度、季度数据及 NI 计算

permno	Source	datadate	csho / cshoq	ajex / ajexq	jdate (Annual)	linkdt (Annual)
15631	Quarterly	2019-06-30	12.851	0.00588236	–	–
15631	Quarterly	2019-09-30	NULL	0.00588236	–	–
15631	Quarterly	2020-06-30	1.108	0.05882360	–	–
15631	Quarterly	2020-09-30	1.108	0.05882360	–	–
15631	Annual	2019-06-30	12.851	0.00588236	2019-10-31	2020-04-15
15631	Annual	2020-06-30	1.108	0.05882360	2020-10-31	2020-04-15

Source / datadate	修复前	修复后
Quarterly / 2020-06-30	$\ln(1.108 \times 0.0588236) - 0 = -2.730656$	$\ln \frac{1.108 \times 0.0588236}{12.851 \times 0.00588236} = -0.148280$
Quarterly / 2020-09-30	$\ln(1.108 \times 0.0588236) - 0 = -2.730656$	NULL
Annual / 2020-06-30	$\ln(1.108 \times 0.0588236) - 0 = -2.730656$	$\ln \frac{1.108 \times 0.0588236}{12.851 \times 0.00588236} = -0.148280$

			ni	
date	jdate	permno	Before	After
2020-12-31	2020-11-30	15631	-2.730656	-0.148280
2021-01-31	2020-12-31	15631	-2.730656	-0.148280
2021-02-28	2021-01-31	15631	-2.730656	-0.148280
2021-03-31	2021-02-28	15631	-2.730656	-0.148280
2021-04-30	2021-03-31	15631	-2.730656	-0.148280

五、CINVEST

同一公司、连续季度

按 GHZ 季度公式，用 JKP 方法匹配同一公司、对应季度。

修复前 7.6885%（66,323 / 862,631） **修复后** 0%（0 / 798,038）
代码 `chars_ciz/accounting.py` **参考** GHZ（季度公式、代码至少五条记录）；JKP（公司及月份匹配）

原代码

```
.shift(k).over('permno')
```

修复代码（同公司、连续五季）

```
calendar_lag(data_rawq, c, k, 3, history=cinvest_history)
```

GHZ 作者稿，PDF 39：CINVEST 季度公式（非已核实发表版）

cinvest	Titman, Wei & Xie	2004, JFQA	Change over one quarter in net PP&E (*ppentq*) divided by sales (*saleq*) - average of this variable for prior 3 quarters; if *saleq* = 0, then scale by 0.01.

JKP `accounting_chars.sas`，第 133–147 行：先补齐日期，再连接原财报

```
        /* First ensure that the gap between two dates is always one month */
        proc sql;
                create table __comp_dates1 as
                select gvkey, curcd, min(datadate) as start_date, max(datadate) as end_date
                from &data.
                group by gvkey, curcd;
        quit;

        %expand(data=__comp_dates1, out=__comp_dates2, id_vars=gvkey, start_date=start_date,
        ↪  end_date=end_date, freq='month', new_date_name=datadate);

        proc sql;
                create table __helpers1 as
                select a.gvkey, a.curcd, a.datadate, b.*, not missing(b.gvkey) as data_available
                from __comp_dates2 as a
                left join &data. as b on a.gvkey=b.gvkey and a.curcd=b.curcd and a.datadate=b.datadate;
```

$$\text{CINVEST}_q = d_q - \text{mean}(d_{q-1}, d_{q-2}, d_{q-3}).$$

GHZ：“Change over one quarter in net PP&E (ppentq) divided by sales (saleq)”，再减前三季该比值的均值；“if saleq = 0, then scale by 0.01.”（作者稿表 1）

HXZ：年度 CAPX/SALE 除以前三年均值，再减 1；“We exclude firms with sales less than 10 million dollars.”（A.3.1）

permno=12350：旧公司 gvkey=163806，新公司 gvkey=040116

	ppentq		(ppentq - ppentq_l1) / 0.01	
datadate	gvkey=163806	gvkey=040116	Before	After
2021-03-31	0.767	无记录	–	–
2021-06-30	0.732	无记录	$(0.732-0.767)/0.01=-3.5$	–
2021-09-30	–	无记录	–	–
2021-12-31	–	15.317	$(15.317-0.732)/0.01=1458.5$	NULL：缺 2021-09
2022-03-31	–	15.120	$(15.120-15.317)/0.01=-19.7$	-19.7
2022-06-30	–	15.383	$(15.383-15.120)/0.01=26.3$	26.3
2022-09-30	–	19.401	$(19.401-15.383)/0.01=401.8$	401.8
2022-12-31	–	94.561	$(94.561-19.401)/0.01=7516$	7516

	cinvest	
datadate	Before	After
2022-06-30	$26.3-(-19.7+1458.5-3.5)/3=-452.133333$	NULL：仅 3/5 季，缺 2021-06、09
2022-09-30	$401.8-(26.3-19.7+1458.5)/3=-86.566667$	NULL：仅 4/5 季，缺 2021-09
2022-12-31	$7516-(401.8+26.3-19.7)/3=7379.866667$	7379.866667（5/5 季齐全）

NULL：新公司历史不足。销售额为零时按 GHZ 用 0.01 作分母。

					cinvest	
date	jdate	permno	使用季度	已有季度 / 5	Before	After
2023-01-31	2022-12-31	12350	2022-06-30	3/5	-452.133333	NULL
2023-02-28	2023-01-31	12350	2022-09-30	4/5	-86.566667	NULL
2023-03-31	2023-02-28	12350	2022-09-30	4/5	-86.566667	NULL
2023-04-30	2023-03-31	12350	2022-09-30	4/5	-86.566667	NULL
2023-05-31	2023-04-30	12350	2022-12-31	5/5	7379.866667	7379.866667

六、ABR

同一公告重复连接日收益

市场收益用 CRSP 市值加权 `vwretd`（HXZ A.1.2）。

修复前 0.1723%（4,378 / 2,540,220） **修复后** 0%（0 / 1,025,121）
代码 `chars_ciz/abr.py` **参考** HXZ（四日窗口、季末后六个月内）

原代码

```
con.register("ccm_data", ccm1.to_arrow())
df.filter(pl.col("abrd").is_not_null())
```

修复代码

```
ccm1.unique(subset=["permno", "rdq_trad"],
↪  keep="first")
pl.when(pl.col("abrd").count() ==
↪  4).then(pl.col("abrd").sum())
```

HXZ A.1.2，p.2072（PDF 54）：ABR 为公告前两日至后一个交易日

A.1.2 Abr1, Abr6, and Abr12, cumulative abnormal returns around earnings announcement dates. We calculate cumulative abnormal stock return (Abr) around the latest quarterly earnings announcement date (Compustat quarterly item RDQ) (Chan, Jegadeesh, and Lakonishok 1996):

$$\text{Abr}_i = \sum_{d=-2}^{+1} r_{id} - r_{md}, \tag{A1}$$

in which r_{id} is stock i's return on day d (with the earnings announced on day 0) and r_{md} is the value-weighted market index return. We cumulate returns until one (trading) day after the announcement date to account for the 1-day-delayed reaction to earnings news.

$$\mathrm{ABR} = \sum_{k=-2}^{1} (\mathrm{ret}_k - \mathrm{mkt}_k).$$

公告信息超过 `datadate+6mo` 不再沿用。

同一股票、同一公告日的两条财年标签记录只计一次（SEC 8-K）。

`comp_fundq.parquet`

记录	gvkey	permno	datadate	rdq	fyearq / fqtr	ibq
A1	012181	10200	2011-03-31	2011-06-09	2010 / 4	-2.031
A2	012181	10200	2011-03-31	2011-06-09	2011 / 1	-2.031
B1	012181	10200	2011-06-30	2011-07-27	2011 / 1	-0.056
B2	012181	10200	2011-06-30	2011-07-27	2011 / 2	-0.056

两条记录使日收益重复、窗口编号错位；修复后按唯一公告计数，红色为计入窗口的收益。

rdq=2011-06-09

				count	
permno	date	记录	abrd	Before	After
10200	2011-06-07	A1	-0.002188	−4	−2
10200	2011-06-07	A2	-0.002188	−3	
10200	2011-06-08	A1	0.017529	−2	−1
10200	2011-06-08	A2	0.017529	−1	
10200	2011-06-09	A1	-0.032800	0	0
10200	2011-06-09	A2	-0.032800	1	
10200	2011-06-10	A1	0.013700	2	1
10200	2011-06-10	A2	0.013700	3	
10200	ABR			-0.030542	-0.003759

rdq=2011-07-27

				count	
permno	date	记录	abrd	Before	After
10200	2011-07-25	B1	0.004097	−4	−2
10200	2011-07-25	B2	0.004097	−3	
10200	2011-07-26	B1	0.019605	−2	−1
10200	2011-07-26	B2	0.019605	−1	
10200	2011-07-27	B1	0.014824	0	0
10200	2011-07-27	B2	0.014824	1	
10200	2011-07-28	B1	-0.026470	2	1
10200	2011-07-28	B2	-0.026470	3	
10200	ABR			0.068858	0.012056

				abr	
permno	date	jdate	rdq	Before	After
10200	2011-05-31	2011-04-30	2011-02-09	-0.081002	-0.081002
10200	2011-06-30	2011-05-31	2011-02-09	-0.081002	-0.081002
10200	2011-07-31	2011-06-30	2011-06-09	-0.030542	-0.003759
10200	2011-08-31	2011-07-31	2011-07-27	0.068858	0.012056
10200	2011-09-30	2011-08-31	2011-07-27	0.068858	0.012056

七、RET

形成月与下一月收益匹配

修复前 0.0852%（3,152 / 3,700,512） **修复后** 0%（0 / 3,699,949）
代码 `chars_ciz/impute_rank_output.py` **参考** HXZ（2020）§2.1：月末形成组合，使用下一月收益。

原代码

```
pl.col('ret').shift(-1).over('permno')
```

修复代码

```
.with_columns(pl.col('jdate').dt.offset_by('1mo').dt.month_end().alias('date'))
.join(returns, on=['permno','date'], how='left', validate='m:1')
```

HXZ §2.1，p.2025（PDF 7）：年度六月形成；季度非盈利信息滞后四个月

For annually sorted deciles, we split stocks at the end of June of each year t into deciles on a variable measured at the fiscal year ending in calendar year $t-1$ and calculate decile returns from July of year t to June of $t+1$. For monthly sorted portfolios involving the latest earnings data, we use quarterly earnings data in the months immediately after quarterly earnings announcement dates (Compustat quarterly item RDQ). For monthly sorted portfolios involving quarterly accounting data other than earnings, we impose a 4-month lag between the fiscal quarter end and subsequent returns. Unlike earnings, other quarterly

		Before		After	
permno	jdate	date	ret	date	ret
10028	2012-03-31	2012-11-30	-0.036581	2012-04-30	-0.096257
10051	2016-01-31	2018-09-30	-0.119239	2016-02-29	-0.045219
10353	2010-07-31	2015-07-31	0.042781	2010-08-31	-0.111475

八、age

由财报条数改为历史覆盖月数

Before 有效年度财报条数 **After** 数据库覆盖的日历月数
代码 `chars_ciz/accounting.py`；`functions.py`（`add_age`）
参考 JKP：firm_age，在月度公司匹配后独立计算。

$$t_0 = \min\{s_{\text{CRSP}}(\text{permco}),\ s_{\text{Acc}}(\text{gvkey}),\ s_{\text{ID}}(\text{permno})\},$$

$$\text{age}_t = 12\,[\text{year}(t) - \text{year}(t_0)] + \text{month}(t) - \text{month}(t_0).$$

age 衡量可追溯的公司/证券历史。JKP 另用 `comp_ret_first`；本地无 `comp_secm/comp_g_secd`，以 permno 首月代替。

原代码

```
pl.col('count').alias('age')
```

修复代码

```
first_date = min(crsp_first, comp_acc_first, first_alt)
age = 12 * (jdate.year - first_date.year) + jdate.month - first_date.month
```

JKP `aux_functions.py`，第 7197–7206 行：age 为覆盖月数

```
def firm_age(paths: DataPaths, data_path):
    """
    Description:
        Compute firm age in months using earliest of CRSP, Compustat accounting, or Compustat returns dates.

    Steps:
        1) Load identifiers/dates from inputs; get earliest dates per gvkey/permco.
        2) Join earliest sources to each (id, eom); also get first observed eom per id.
        3) Age = months between eom and min(first_obs, first_alt). Write result.
```

财报条数改为覆盖月数

数据文件均为 `chars_raw_no_impute.parquet`；Before：`data/processed`；After：`data/processed_20261030`。

date	jdate	permno	Before（年频计数）	After（月）
2023-01-31	2022-12-31	10026	37	456
2023-02-28	2023-01-31	10026	20	457
2023-03-31	2023-02-28	10026	20	458
2023-04-30	2023-03-31	10026	20	459
2023-05-31	2023-04-30	10026	20	460

九、通用：公司归属

公司切换：不能延用旧公司财报

修复前 0.4936%（15,479 / 3,136,020） **修复后** 0%（0 / 3,120,541）
代码 chars_ciz/accounting.py；chars_ciz/impute_rank_output.py **参考** JKP（按月检查 CCM 有效链接）

Ref	查得的处理
HXZ	未明确规定 gvkey 切换后是否延用旧财报。
GHZ	财报截止日检查 CCM 后延用；未逐月核对公司，可能跨越切换日。
CZ	财报截止日检查 CCM，滞后六个月并延用十二个月；可能跨越切换日。
JKP	按股票当月日期检查 CCM，再按当月 gvkey、月份匹配；采用此法。

原代码

```
pl.all().forward_fill().over(['permno', 'datadate'])
```

修复代码

```
(pl.col('linkdt').is_null() | (pl.col('linkdt') <= pl.col('mthcaldt'))) &
(pl.col('linkenddt').is_null() | (pl.col('mthcaldt') <= pl.col('linkenddt')))
pl.when(pl.col('gvkey') == pl.col('_current_gvkey')).then(pl.col(v)).alias(v)
```

JKP aux_functions.py，**第 603–615 行：以股票日期检查 CCM 链接**

```
    full_join = sf_senames_join.join(
        ccmxpf_lnkhist,
        how="left",
        predicates=[
            (sf.permno == ccmxpf_lnkhist.lpermno),
            ((date_expr >= ccmxpf_lnkhist.linkdt.cast("date")) | ccmxpf_lnkhist.linkdt.isnull()),
            (
                (date_expr <= ccmxpf_lnkhist.linkenddt.cast("date"))
                | ccmxpf_lnkhist.linkenddt.isnull()
            ),
            ccmxpf_lnkhist.linktype.isin(["LC", "LU", "LS"]),
        ],
    )
```

$$\text{ADM} = \frac{\texttt{xad}}{\texttt{me}}.$$

新公司 2021 年 xad 缺失，ADM 置空；切换月 gvkey 由 163806 改为 040116。

本表直接读取两份最终未填补数据：Before 为 data/processed，After 为 data/processed_20261030，不重算任一侧。旧 age 是年度财报累计记录数（年频计数），新 age 是历史覆盖月数，不能直接比较其大小。

			Before			After		
permno	date	jdate	gvkey	age（年频计数）	adm	gvkey	age（月）	adm
12350	2021-11-30	2021-10-31	163806	3	0.003406	163806	142	0.003406
12350	2021-12-31	2021-11-30	163806	3	0.001203	040116	133	NULL
12350	2022-01-31	2021-12-31	163806	3	0.001781	040116	134	NULL
12350	2022-02-28	2022-01-31	163806	3	0.002590	040116	135	NULL
12350	2022-03-31	2022-02-28	163806	3	0.003776	040116	136	NULL
12350	2022-04-30	2022-03-31	163806	3	0.000258	040116	137	NULL
12350	2022-05-31	2022-04-30	163806	3	0.000583	040116	138	NULL
12350	2023-02-28	2023-01-31	040116	2	0.007568	040116	147	NULL
12350	2023-03-31	2023-02-28	040116	2	0.010874	040116	148	NULL